

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2015

BIOLOGY

HIGHER 2

9648/02
24 AUGUST 2015
2 hours

Paper 2 Core Paper

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
9 or 10	
Total	100

This question paper consists of **20** printed pages.

- 1 (a) Fig. 1.1 is a schematic diagram showing the transport pathways of extracellular and intracellular materials for digestion in a mammalian cell. Depending on the types of digested material, three possible pathways are initiated to deliver these materials for digestion within lysosomes, of which two are labelled as **A** and **B**.

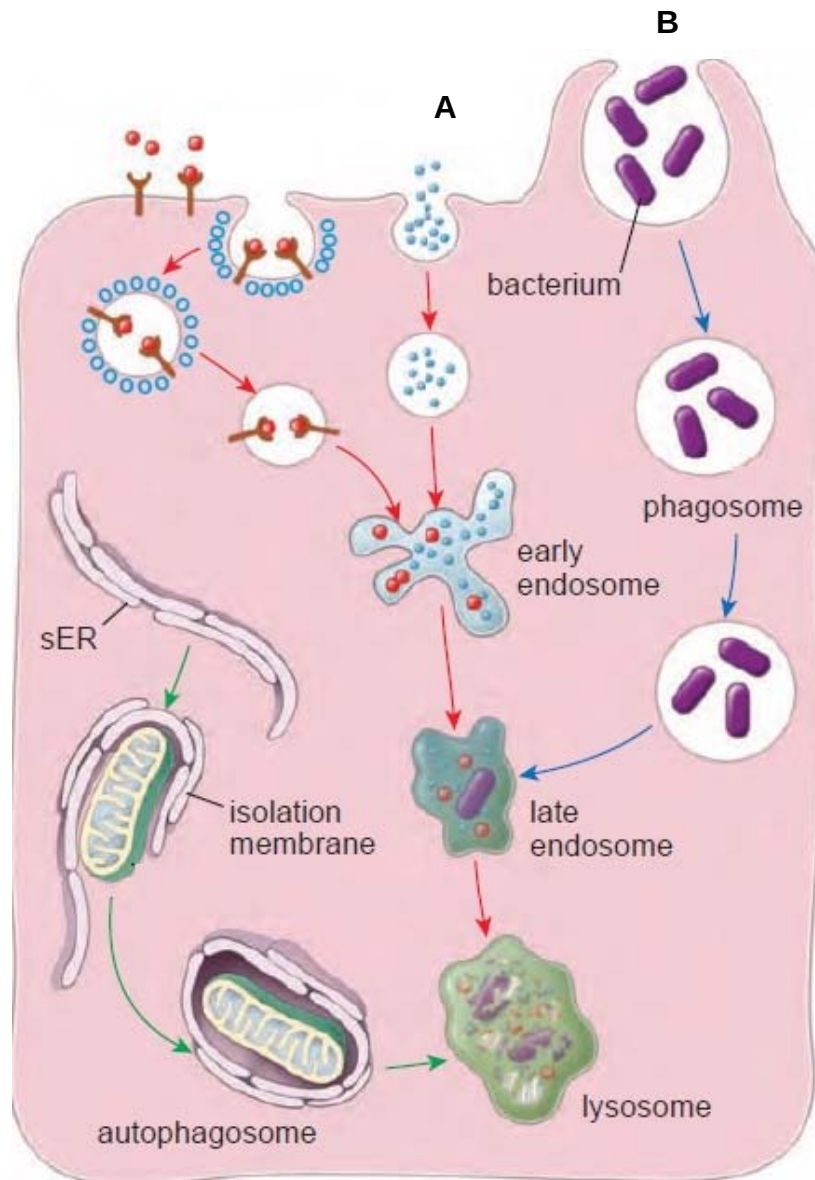


Fig. 1.1

- (i) State **one** property of the plasma membrane and explain how it enables process **A** to be carried out by a cell. [2]

1. Membrane is **fluid**;

2. Ref. to invagination of plasma membrane/ fusion of two ends of plasma membrane to form endocytic vesicle;

@ 1m

- (ii) Degradation of worn out organelles such as mitochondria occurs inside most cells via autophagy. With reference to Fig. 1.1, describe the process of autophagy. [3]

1. Isolation membrane (derived from SER) encloses mitochondria to form a (membrane bound) autophagosome;
2. Fusion of membrane of autophagosome with membrane of lysosome;
3. Hydrolytic enzymes released into autophagosome; R! digestive enzymes
4. Digestion of mitochondria by hydrolytic enzymes, (which are activated by low pH within lysosome);

@ 1m

- (iii) Process B requires rapid rearrangement of the cytoskeleton when extracellular materials bind to cell surface membrane receptors. This rapid rearrangement requires the activation of a variety of protein kinases which occurs when there is a surge in the cytosolic levels of calcium ions. Explain the role of calcium ions to bring about this cellular response. [3]

1. Calcium ions act as second messengers which relay/transduce external signal to inside the cell;
2. by binding to and activating protein kinases, resulting in phosphorylation cascade;
3. (Activated protein kinases bind to) effector proteins, (resulting in change in 3D configuration) and hence activated (to bring about rearrangement of cytoskeleton);
4. Ref. signal amplification to bring about rearrangement of cytoskeleton rapidly;

@ 1m, max 3

- (b) Lysosomes are rich in enzymes such as proteases and nucleases. The lysosomal membrane contains an unusually large amount of integral proteins, which are highly glycosylated with oligosaccharides on the surface of membrane facing the interior.

- (i) Briefly describe how an oligosaccharide is formed from glucose. [3]

1. These glucose molecules are linked by glycosidic bonds R! covalent bonds

2. during condensation reaction (with removal of water molecules);

3. Ref. to catalysis by enzymes; @ 1m

- (ii) Suggest the role of these glycosylated proteins in lysosomes. [1]

1. Ref. to large amounts of these glycoproteins acting as a molecular shield/ OWTTE;
2. Hindering the access / contact of enzymes (e.g. proteases, phospholipases) with substrates (e.g. membrane protein, phospholipids) / preventing the binding of the substrates to the active sites of enzymes;
3. Ref. to prevent self - digestion of lysosome by hydrolytic enzymes/ OWTTE;

@ 1m, max 1

[Total: 12m]

- 2 Transglutaminase (TGase) secreted by *Streptomyces mobaraense* is a valuable enzyme for the food industry. TGase, better known to chefs as “meat glue”, has the ability to bond protein-containing foods together. Some of its uses include the production of restructured meat and the hardening of fish protein paste.

TGase consists of 331 amino acids, including only one cysteine residue. The three amino acid residues at the catalytic site are cysteine 64, aspartate 255 and histidine 274, where the numbers denote the positions of the amino acids in the amino acid sequence of TGase. Fig. 2.1 shows the structure of TGase.

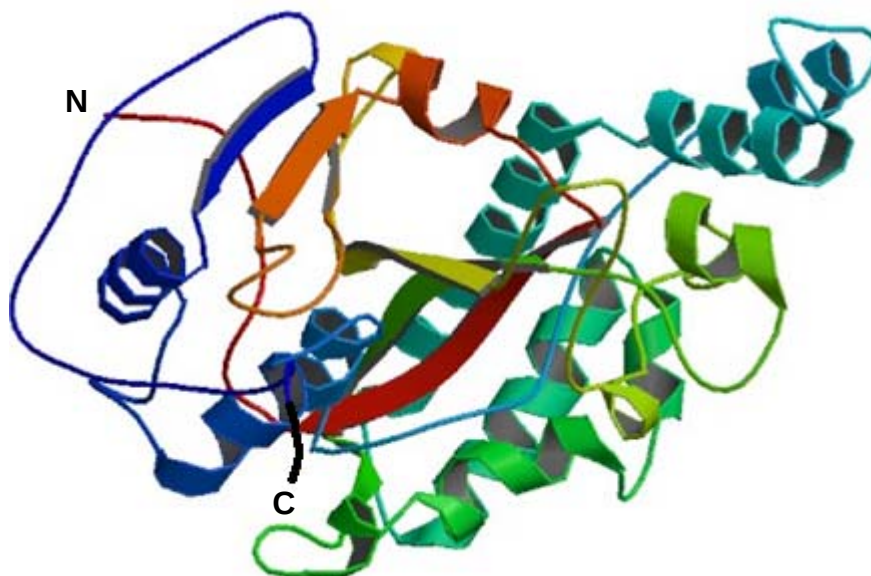


Fig. 2.1

- (a) With reference to Fig. 2.1 and your knowledge of haemoglobin, state **two** structural differences between TGase and haemoglobin. [2]

TGase	Haemoglobin
1. Consist of <u>α-helices and β-pleated sheets</u>	Consist of only <u>α-helices;</u>
2. Highest level of structure is <u>tertiary structure/ No quaternary structure</u>	Highest level of structure is <u>quaternary structure/ quaternary structure;</u>
3. Consists of only <u>one polypeptide chain/ No dimer</u>	Consists of <u>four polypeptide chains/ 2 identical dimers;</u>
4. Does not carry a <u>prosthetic/ haem group</u>	Carries a <u>prosthetic/ haem group</u> containing Fe^{2+} ;
5. Consists of a <u>catalytic site</u>	Does <u>not</u> consist of <u>catalytic site;</u>
6. Consists of <u>one cysteine</u>	Consists of <u>many cysteines;</u> R! if only reference to one subunit
7. Consists of 331 aa	Consists of 141 aa (alpha chains) & 146 (beta chains);

@ 1m, max 2

- (b) Explain what determines the precise position of the three catalytic residues in the active site of TGase. [2]

1. Determined by the primary structure of the protein, which is the specific number and sequence of amino acid along the polypeptide chain;
2. The formation of hydrogen bonds, ionic bonds and hydrophobic interactions (name any 2) between R groups of amino acids in the polypeptide chain; (R! disulphide as there is only 1 cysteine residue in the polypeptide chain)
3. bringing the three distantly separated amino acids into close proximity, giving the enzyme its tertiary structure / three-dimensional conformation brings these amino acids into close proximity; @1m, max 2

- (c) Describe the mode of action of TGase. [3]

1. Ref. to three-dimensional conformation of active site is complementary to the protein/ lock and key hypothesis/ induced fit hypothesis, forming enzyme-substrate complexes;
2. Ref. to formation of temporary bonds with active site of enzyme/ Ref. to chemical fit;
3. Enzyme lowers activation energy;
4. Ref to how activation energy is lowered (Proximity effect/ orientation surface/ microenvironment effect/ acid-base catalysis); R! Bond straining
5. Ref. to catalyzing bonds between protein-containing food; @ 1m, max 3

[Total: 7m]

- 3 Fig. 3.1 shows the sequence of nucleotide bases in a section of a viral gene coding for the first seven amino acids of a viral protein, X.

CTTCTGTTTCTGCTGGAACAT

Fig. 3.1

- (a) Methionine is coded by the initiation codon. List the corresponding codons on the mRNA (in the 5' to 3' direction) transcribed from the template in Fig. 3.1. [1]

5' - AUG UUC CAG CAG AAA CAG AAG - 3'

- (b) Protein X was hydrolyzed to release the first seven amino acids which comprises of four different amino acids. The number of each type of amino acid obtained is shown in Table 3.1.

Table 3.1

Amino acid	Number present
Phe	1
Met	1
Lys	2
Gln	3

Using information in Table 3.1, list the first seven amino acids of protein X in order, starting from the amino acid at the N-terminus. [1]

Met, Phe, Gln, Gln, Lys, Gln, Lys

@ 1m

- (c) Explain if the replication of the influenza virus genome can be described as semi-conservative. [3]

1. The replication cannot be described as semi-conservative;

2. The resulting daughter RNA molecule is a single strand;

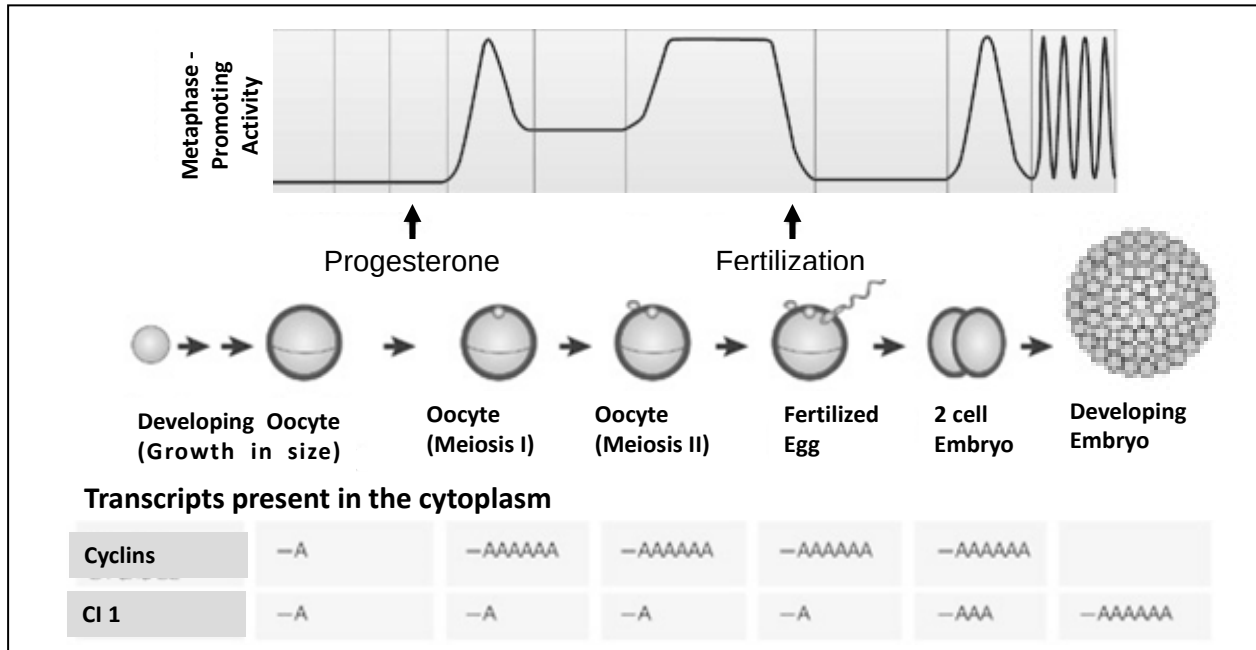
3. Does not consist of one parental strand and one daughter strand/ Semi-conservative replication will result in one parental strand and one daughter strand;

4. Parental RNA does not serve as a direct template for synthesis of daughter RNA (but through an intermediate mRNA);

@ 1m, max 3

[Total: 5m]

- 4 Fig. 4.1 shows a series of events that occur during the maturation of an oocyte leading to its fertilization and the development of the subsequent embryo. The corresponding metaphase-promoting activity due to proteins present in the various stages of the oocyte and the embryo are shown in a graph above the cells. The mRNA transcripts of cyclins and Cl 1 with their corresponding length of poly-A tails at various stages are shown below the cells.



Legend	
A	Short poly-A tail
AAA	Mid length poly-A tail
AAAAAAA	Long poly-A tail

Fig. 4.1

- (a) With reference to Fig. 4.1, state and explain the mode of control of gene expression that leads to the maturation of the oocyte until meiosis II. [4]

1. **Translational control of protein;**
2. **Shown by the changes in the length of the poly A tail of cyclin mRNAs induced by progesterone;**
3. **A longer poly A tail will initiate translation / A shorter poly A tail will prevent initiation of translation;**
4. **Longer poly A tail increases the half-life of the mRNA transcript allowing longer time for translation/ OWTTE;**
5. **High amount of cyclins produced allow development of oocyte;**

@1m, max 4

- (b) Suggest the advantage of the type of gene control that is present in the early stages of the embryo. [2]

1. **Allow rapid changes in the protein levels in the embryo in response to fertilization;**
2. **To bring about rapid development in the stages of the embryo;**
3. **More energy efficient;**

@1m, max 2

(c) Cyclin is a protein that can be found in most cells in the body. Explain how a mutation in the gene coding for cyclin could contribute to the development of cancer. [4]

1. Cyclin is coded by proto-oncogene as it promotes (metaphase in) mitosis/meiosis;
2. As shown by its poly adenylation level correlating with the level of metaphase-promoting activity;
3. Gain of function mutation causes cyclin to be hyperactive / resistance to degradation/ high expression of cyclin;
4. Cell **may** bypass check points in mitotic cell cycle, **leading to accumulation of mutations leading to angiogenesis / metastasis**;
5. Leading to uncontrolled cell division/ **cell divide even in the absence of chemical / growth signal**;

@1m, max 4

[Total: 10m]

- 5 (a) Recent findings suggest that the mucus lining the human digestive tract is loaded with bacteriophages. These bacteriophages contain proteins which bind to the carbohydrate residues of mucoproteins as shown in Fig. 5.1. It has been proposed that these bacteriophages can play an important role as part of the body's defense system.

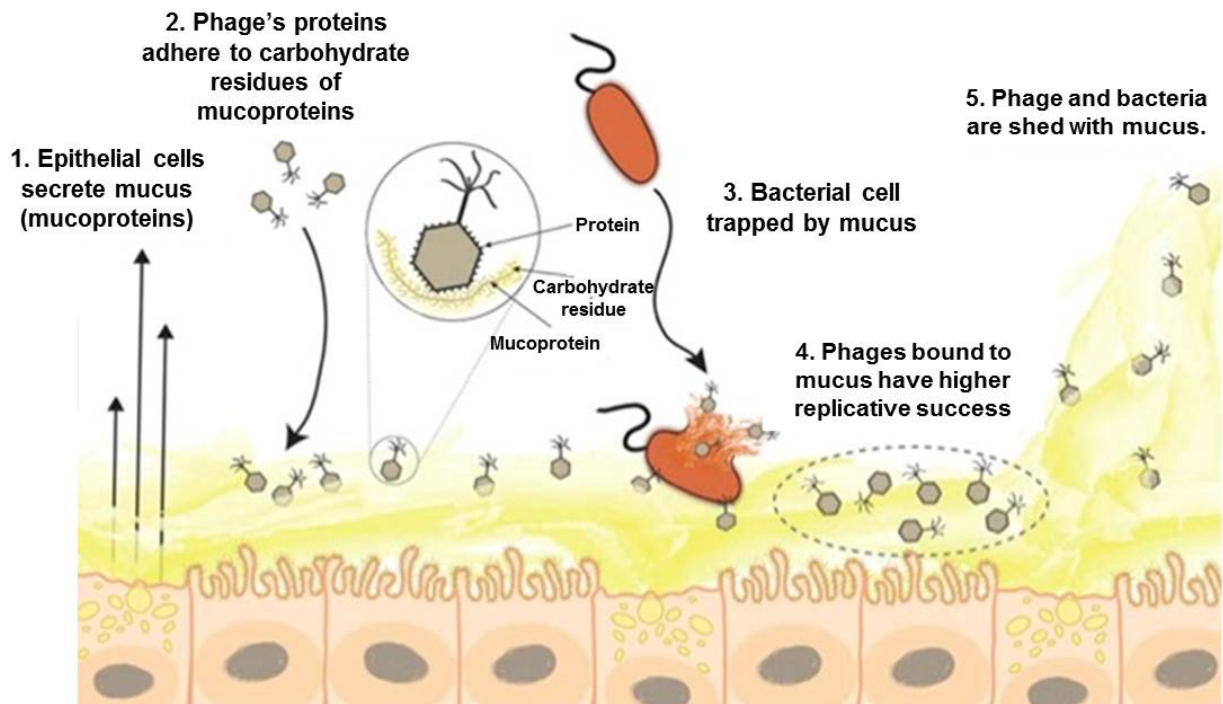


Fig. 5.1

- (i) Describe the life cycle of the bacteriophages shown in Fig. 5.1. [4]

1. The bacteriophages use their tail fibres to bind to specific receptors on host bacteria;
 2. Phage-coded enzyme, lysozyme, digests part of the cell wall of host bacterium/ and viral DNA/ genome is injected;
 3. Ref. to transcription and translation/ expression of phage proteins by using host RNA polymerase and ribosomes;
 4. Ref. to replication of phage DNA using host DNA polymerase;
 5. Newly synthesized bacteriophage proteins and DNA assembled into new phages and are released from the bacteria by osmotic lysis, killing bacteria in the process;
- @1m, max 4

- (ii) With reference to Fig. 5.1, suggest how the way in which bacteriophages bind to mucoproteins is significant to their higher replicative success. [2]

1. Capsid heads bind to mucoproteins;
2. Bacteriophages' tail fibres faced upwards so that they can bind to specific receptors on host bacteria;
- *3. Hence, higher efficiency of bacteriophages/ **high concentration of bacteriophage to accumulate in the mucus, hence** infecting the host bacteria for replication/;

*Valid only when Point 1/2 present; @1m

- (iii) Many other viruses may also be found within the human body e.g. the Human Immunodeficiency Virus, HIV. Compare between the lysogenic life cycle of a temperate bacteriophage and life cycle of a HIV. [3]

Similarity:

1. Both life cycles involve integration of viral DNA into host cell genome;
2. Both life cycles involve a period of dormancy;
3. Both consist of proteins which bind to specific host cell receptors;

Lysogenic life cycle	HIV life cycle
4. Use tail tip to bind to specific receptors on bacterial cells	Use gp120 to bind to CD4 receptors on Helper T cells;
5. Only DNA is injected into host cell /Capsid left outside host cell	Capsid and viral RNA enters host cell;
6. Viral DNA integrated into host cell genome directly	Viral RNA needs to be reversed transcribed to viral DNA before integration into host cell genome;
7. Occurs in host cell bacteria	Occurs in T helper cells/ cells with CD4 receptors;
8. Do not result in death of the host cell	HIV may result in death of host cell;

At least 1 similarity and 1 difference @1m, max 3

- (b) In bacteria, genes coding for enzymes involved in the same metabolic pathway are arranged into operons. Fig. 5.2 shows the changes in the activity of enzymes that synthesise tryptophan and utilise lactose in a cell after the addition of tryptophan and lactose.

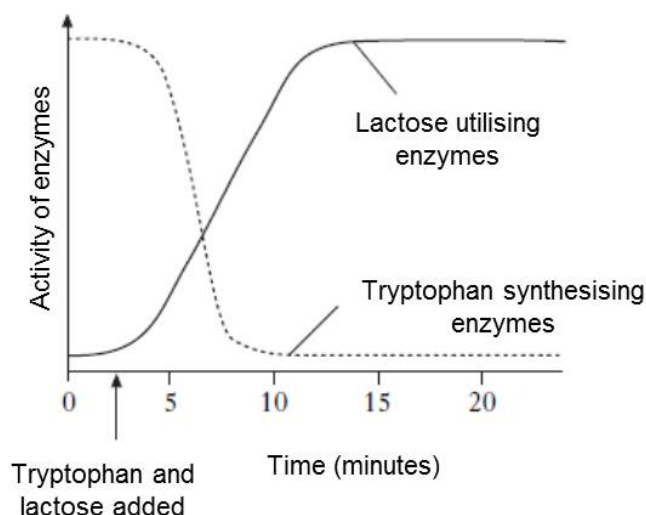


Fig. 5.2

A scientist altered the *trp* operator sequence *in vitro* such that it was no longer complementary to the binding sites of repressor proteins. The scientist also altered the *lac* repressor gene sequence *in vitro* such that the repressor binding site was no longer complementary to its inducer.

Describe and explain the effects of the mutations on the activity of lactose utilising enzymes and tryptophan synthesizing enzymes upon addition of lactose and tryptophan. [4]

- *1. Lactose utilization enzymes remain low and tryptophan synthesizing enzymes remain high;
2. Allolactose is not able to bind to *lac* repressor due to altered binding site, *lac* repressor remains active/ binds to operator;
3. RNA polymerase cannot bind to promoter, hence no transcription and translation/ expression of *lac* structural genes;
4. *trp* repressor cannot bind to operator even in the presence of tryptophan;
5. RNA polymerase can bind to promoter, hence transcription and translation/ expression of *trp* structural genes occurs; *compulsory point @ 1m

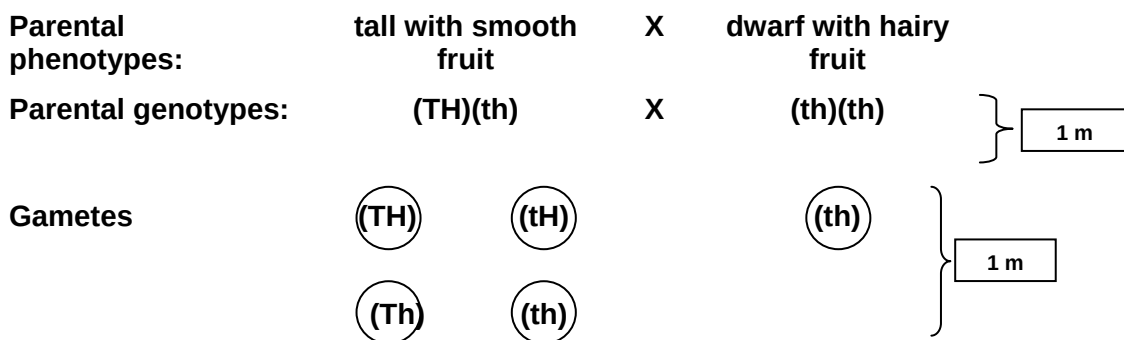
[Total: 13m]

- 6 Analysis of genes on separate loci in tomato plants were carried out. In the experiment, pure-breeding tall tomato plants with smooth fruits were crossed with pure-breeding dwarf tomato plants with hairy fruits. All the plants in the F_1 generation were tall with smooth fruits.

When a F_1 plant was subjected to a test cross, the following offspring phenotypes were observed:

Tall with smooth fruit	80
Tall with hairy fruit	20
Dwarf with smooth fruit	19
Dwarf with hairy fruit	81

- (a) Using the symbols **T** and **t** for height and **H** and **h** for fruit type, draw a genetic diagram to explain the observed results of the test cross. [5]



Punnett Square:

	(TH)	(tH)	(Th)	(th)
(th)	(TH)(th) Tall with smooth fruit	(tH)(th) Dwarf with smooth fruit	(Th)(th) Tall with hairy fruit	(th)(th) Dwarf with hairy fruit

1 m for
genotypes
and 1m for
correct
phenotypes

Indicating recombinant phenotypes/gametes

1m

@1m

(b) (i) State the expected dihybrid test cross ratio.

[1]

1:1:1:1 @1m

(ii) Explain why the observed test cross results deviate from the ratio stated in (b)(i). [3]

- 1. Linked genes; A! incomplete linkage R! complete linkage**
- 2. Both genes do not segregate independently/ Both genes are inherited together;**
- 3. Crossing over to form recombinant gametes is a chance event;**
- 4. Thus resulting in low numbers of recombinant phenotypes compared to parental phenotypes;**

@1m, max 3

A chi-squared test was used to find out if there is any significant difference between the observed and expected ratio.

The formula for χ^2 and the table of probabilities are given as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

where Σ = 'sum of...'

O = observed 'value'

E = expected 'value'

v = degrees of freedom

c = number of classes

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(c) Calculate the χ^2 value and conclude the results of the chi-squared test at 5% level of significance. [3]

1. Calculated χ^2 value = (18 + 19.22 + 18 + 19.22) = 74.4;

2. DOF = 3, Calculated χ^2 value > Critical χ^2 value of 7.82;

3. $p < 0.05$ or $p < 0.001$. Therefore, there is significant difference between the observed and expected ratio and any deviation from the expected ratio is not due to chance;

@1m

[Total: 12m]

- 7 (a) Fig. 7.1 is an electron micrograph of a mitochondrion from an animal cell.

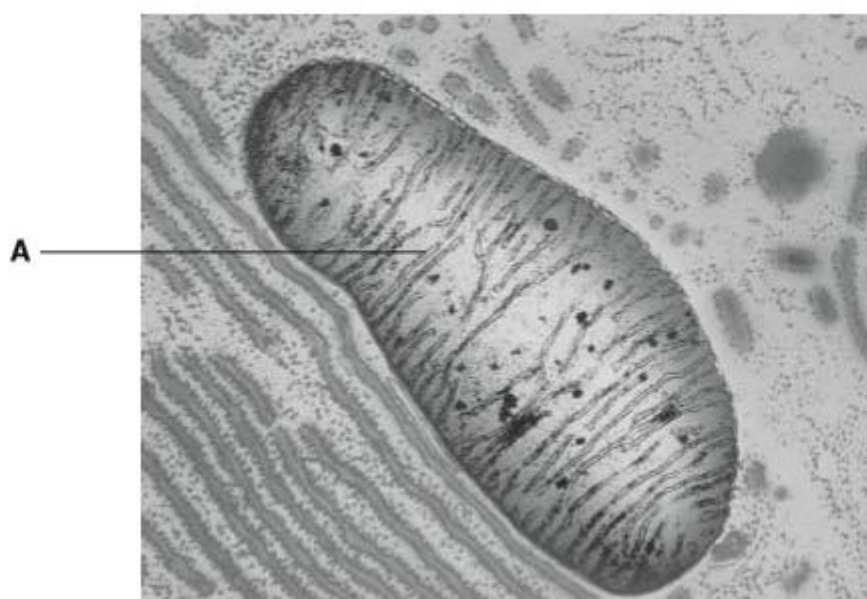


Fig. 7.1

- (i) Name the specific process that is carried out by structure **A** in the mitochondrion. [1]

Oxidative phosphorylation/ chemiosmosis;

@ 1m

- (ii) State **three** differences between the process of ATP formation in structure **A** and in the cytoplasm. [3]

1. Oxidative phosphorylation	Substrate-level phosphorylation;
2. Involves NAD^+ and FAD (as hydrogen carriers)	Does not involve NAD^+ and FAD (as hydrogen carriers);
3. Involves ATP synthase/ electron carriers of the electron transport chain	Does not involve ATP synthase/ electron carriers;
4. Proton gradient generated	Proton gradient is not generated/ energy from a coupled reaction;
5. Yields more (32/34) ATP per glucose	Yields less (net 2) ATP per glucose;
6. Does not occur during anaerobic respiration but only in aerobic respiration/ Oxygen required as a final electron and proton acceptor;	Occurs during anaerobic and aerobic respiration/ Oxygen not required as a final electron and proton acceptor;
7. Does not require expenditure of 2 ATP	Requires expenditure of 2 ATP;

@ 1m, max 3

- (iii) Suggest how the structure of a mitochondrion from a muscle cell would differ from that of a mitochondrion from a skin cell. [1]

1. Bigger in size;

2. Longer/more cristae/ more ATP synthases/ more electron carriers;

@ 1m

- (b) The bacterium, *H. salinarium* (*H. salinarium*) utilises light energy to drive the production of ATP in the cell membrane. Fig. 7.2 shows how the protein, bacteriorhodopsin, present in the cell membrane of *H. salinarium* helps with the process of producing ATP.

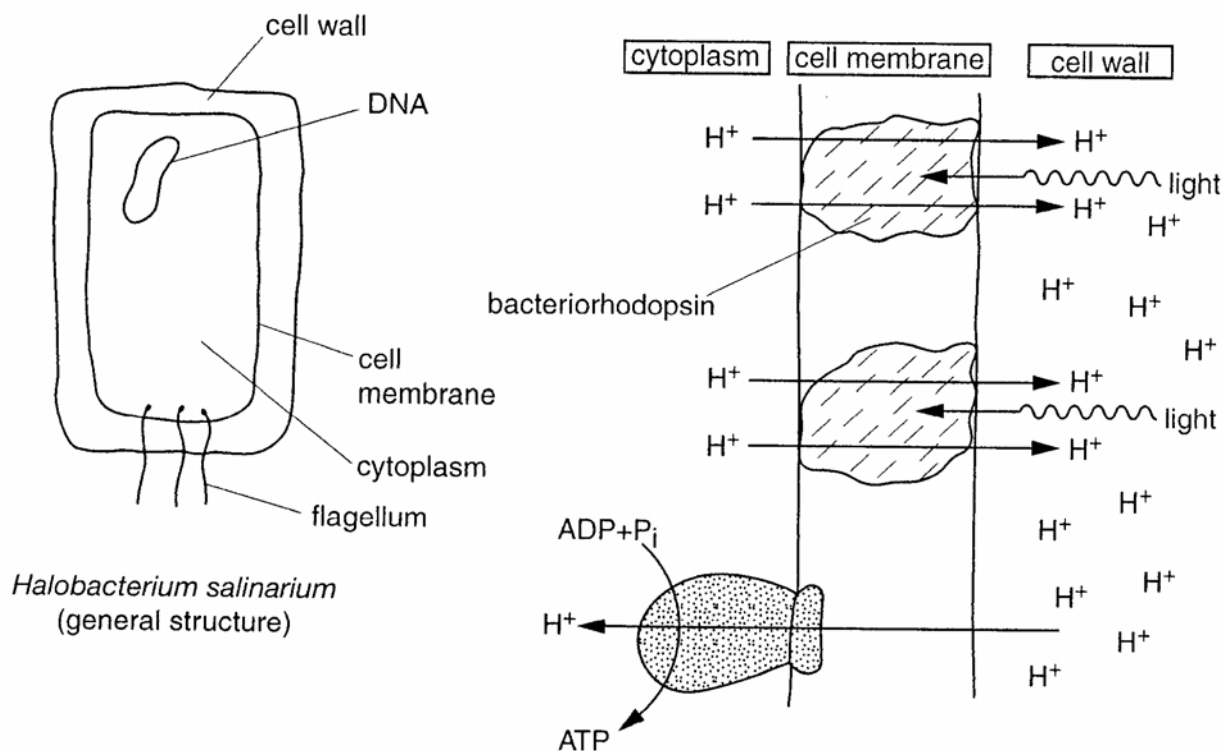


Fig. 7.2

With reference to Fig. 7.2, explain how *H. salinarium* synthesizes ATP.

[4]

1. The bacteriorhodopsin absorbs light energy/ is photo-activated;
 2. which is used to pump protons out of the cytoplasm into the cell wall;
 3. Generating a proton pool in cell wall/proton gradient across the cell membrane;
 4. Chemiosmotic synthesis of ATP via ATP synthases /
- Phosphorylation of ADP to ATP via ATP synthase is powered by the proton motive force/ movement of protons moving down concentration gradient;

@ 1m

[Total: 9m]

- 8 (a) The diagram below represents the various birth weights of new-born calves in a wild population of deer in the African grasslands. The line graph on the diagram represents mortality in relation to birth weight.

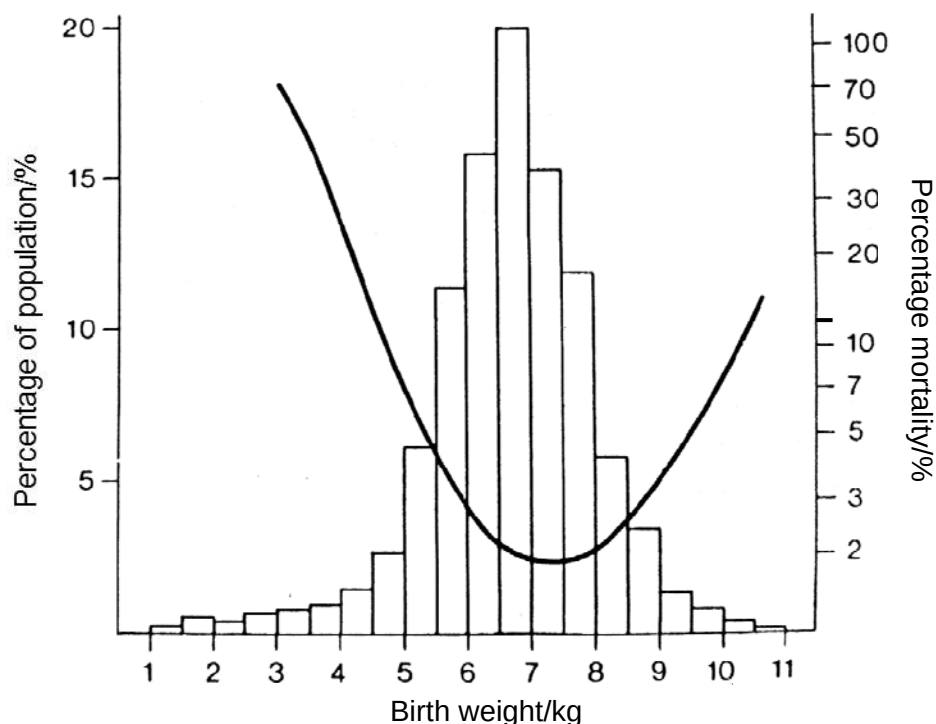


Fig. 8.1

- (i) Using the information provided in Fig. 8.1, account for the type of selection acting on birth weight of new-born deer calves. [4]

1. **Stabilising selection;**

2. Calves with high and low birth weight are selected against/ **lower reproductive success**, hence calves with high and low birth weight have lowest percentage population/ highest mortality;

3. Calves with median birth weight (5 to 8.5kg) are selected for/ **higher reproductive success**, hence calves with median birth weight has highest percentage population/ lowest mortality;

4. Correct values to support point 2 or 3;

@1m

- (ii) Birth weight of new-born deer calves is an example of continuous variation. Explain why there is a variation of birth weights in the population of deer. [3]

1. **Effect of polygenes (birth weights affected by cumulative effects of different genes);**

2. **Additive effects of multiple alleles;**

3. **Ref. to environmental factors;**

4. **Mutations lead to formation of new alleles result in a variation of genotypes;**

5. **During prophase I, crossing over occurs between homologous chromosomes, resulting in gametes with recombinant chromosomes;**

6. **During metaphase I/ II, independent assortment of homologous chromosomes/ chromatids, resulting different combinations of maternal and paternal chromosomes;**

7. **Random fusion of gametes;**

Points 5 to 7 @1m, max 1

- (iii) Suggest why percentage of mortality is higher on both ends of the range of birth weights of new-born deer calves. [2]

1. **Low birth weight → result of premature birth → underdeveloped → higher mortality;**

2. **High birth weight → difficult birth/suffocate in birth canal/heavy and more easily caught by predators;**

3. **AVP;**

@ 1m, max 2

- (b) The deer is an artiodactyl. Artiodactyls are cloven-hooved mammals; they have hooves split into two toes and their weight is borne equally by the two toes. Fig. 8.2 shows a simplified phylogenetic tree of commonly-known artiodactyls and the skeletal anatomy of their hooves.

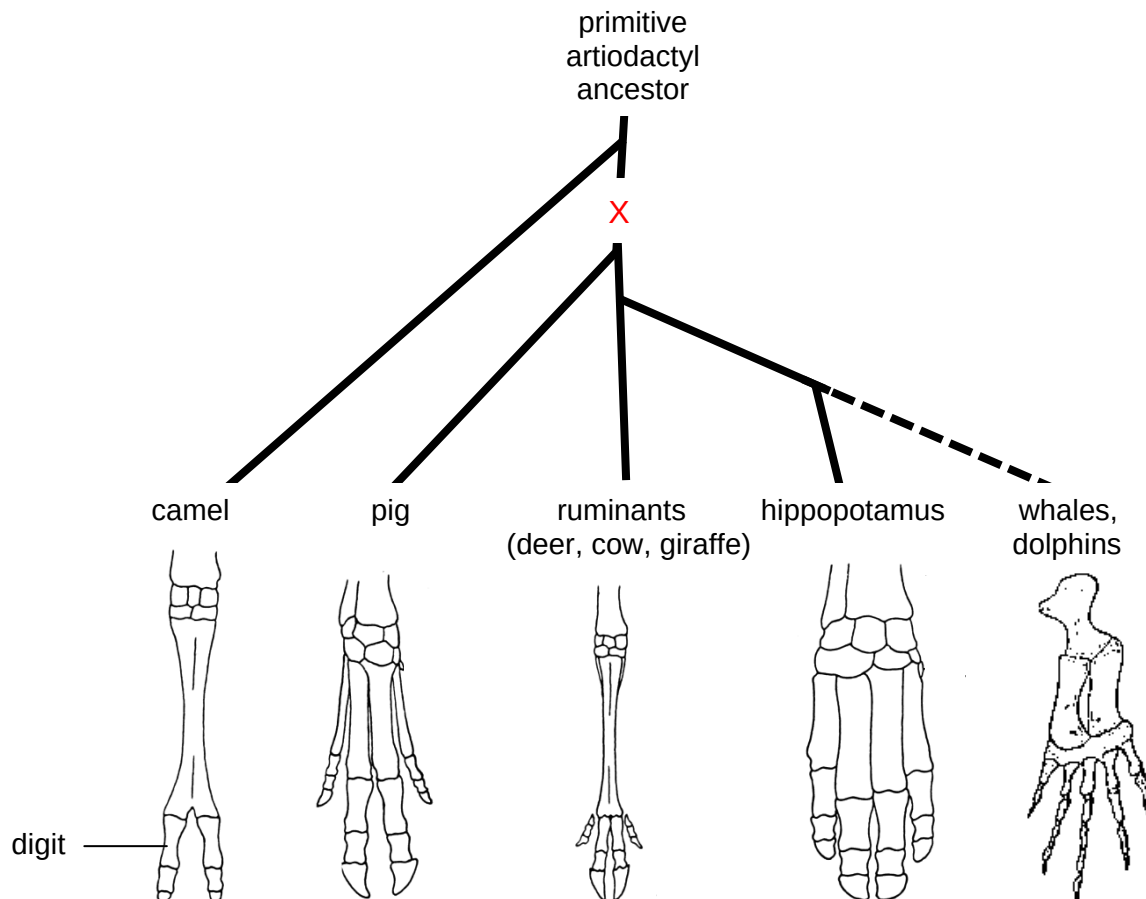


Fig. 8.2

- (i) On Fig. 8.2, use a 'X' to mark where the common ancestor of pigs, ruminants and hippopotamuses could have been positioned on the phylogenetic tree. [1]
- (ii) Recent studies have proposed that the closest relative of aquatic mammals such as whales and dolphins are the hippopotamuses (as indicated by the dotted line in Fig. 8.2). Besides using comparative anatomy, suggest and briefly explain the basis of one other possible approach scientists could have taken to arrive at this proposal. [2]
- 1. Molecular homology/ Ref. to use genes/ amino acid sequences;**
- 2. Organisms which are related would have similar genes/ amino acid sequences as they are derived from a common ancestor;** @1m

[Total: 12m]

Section BAnswer **EITHER 9 OR 10.**

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.**EITHER**

- 9** (a) Outline the differences between facilitated diffusion and exocytosis. [6]
 (b) Describe the process of translation. [8]
 (c) Explain how most evolutionary changes is the result of genetic drift acting on neutral alleles. [6]
- OR**
- 10** (a) Outline the differences between prokaryotic and eukaryotic control of gene expression. [8]
 (b) Explain how a deletion mutation in the gene for DNA polymerase III will prevent the occurrence of DNA replication. [6]
 (c) Explain how biogeography supports the theory of evolution by natural selection. [6]

- END OF PAPER -

9(a) Outline the differences between facilitated diffusion and exocytosis.

[6]

Differences:

	Facilitated diffusion	Exocytosis
1.	Does not require ATP	Requires ATP;
2.	Transports (polar /charged /hydrophilic) small molecules	Transports large macromolecules;
3.	Requires protein carrier /channel to transport substances	Does not require protein carrier/ channel to transport substances;
4.	Does not require microtubules in guiding the transport of substances	Requires microtubules in guiding the movement of vesicle (to cell surface membrane);
5.	Does not involve the fusion of membranes	Involves the fusion of membranes;
6.	Bidirectional movement of substances	Uni-directional movement of substances;
7.	Requires concentration gradient to be established across a membrane	Concentration gradient is not required;
8.	Doesn't add to the cell surface membrane	<u>Addition</u> of cell surface membrane upon fusion;

@1m, max 6

9(b) Describe the process of translation.

[8]

Initiation

1. The specific amino acid is attached to the sequence 3' CCA at end of tRNA;
2. by amino acyl-tRNA synthetase to form aminoacyl-tRNA;
3. Small ribosomal subunit binds to mRNA at the 5' / modified guanosine cap and moves downstream to locate the start codon/ AUG;
4. Aminoacyl-tRNA carrying methionine/ Aminoacyl-tRNA with anticodon UAC (binds to the start codon AUG on mRNA);
5. Large ribosomal subunit then binds and methionine aminoacyl-tRNA is positioned at the P site of the large ribosomal subunit;

Elongation

6. The second aminoacyl-tRNA is held at the A site of the large ribosomal subunit.;
7. The two amino acids are joined by a peptide bond catalysed by peptidyl transferase;
8. And first amino acid (methionine) is now transferred to the aminoacyl-tRNA at the A site, forming peptidyl-tRNA;
9. Ribosome translocates in the 5' to 3' direction, (tRNA originally at the P site is relocated to the E site where it is ejected from the ribosome into the cytoplasm);

Termination

10. When stop codon/ UGA/ UAG/ UAA is reached, release factor protein binds and polypeptide is released from ribosome;
11. Ribosome disassembles into its large and small subunits.;

@1m, max 8

9c) Explain how most evolutionary change at the molecular level is the result of genetic drift of neutral alleles. [6]

1. Neutral alleles can be formed through mutations which occur in non-coding regions/regions coding for non-essential amino acids/result in amino acid with a R-group of similar chemical property/ degeneracy of the genetic code (state any one);
2. Genome largely comprised of non-coding regions, hence most mutations are neutral;
3. No effect on protein function/ reproductive success/ fitness; **A! reverse argument**
4. Hence, neutral alleles do not come under selection pressure; **A! reverse argument**
5. As these variants are neutral, they can be lost or fixed in a population's gene pool;
6. Hence, resulting in a change of allele frequency in a population's gene pool due to chance event;
7. Ref. to an example: Founder's effect /Bottleneck effect;
8. Elaboration of example: Founder's effect in which a new colony is started by individuals from the original population/ Bottleneck effect in which population reduced in size due to sudden event;
9. As these changes remain at the molecular level and accumulate at a constant rate (since mutation rate is considered constant) over the generations, they reflect evolution at the molecular level;

@1m, max 6

10 (a) Outline the differences between prokaryotic and eukaryotic control of gene expression. [8]

Basis	Prokaryotic control of gene expression	Eukaryotic control of gene expression
Arrangement of genes	1. Genes involved in similar metabolic functions are group as an operon and are under the control of a <u>single promoter</u> / genes of the operon are transcribed together into a single mRNA;	Each gene has its own promoter/ Each gene is individually transcribed;
Chromatin remodelling	2. Absence of chromatin remodelling/ absence of histone modification/ DNA modification ;	Presence of chromatin remodelling/ acetylation/ methylation of histones/ presence of histone modification/DNA modification ;
Negative / Positive gene regulation	3. Uses mostly <u>negative</u> regulation (i.e. repressors that bind to operators);	Uses mostly <u>positive</u> regulation because of DNA is stored as chromatin. (This renders most promoters inaccessible, and so genes are normally silent in the absence of other regulation.);
	4. Control of gene expression by enhancers is rare;	Control of gene expression by enhancers is common;
Alternative splicing	5. Introns are rare, hence absence of alternative splicing;	Presence of introns, hence alternative splicing is possible;
Post-transcriptional modification	6. Does not occur;	Addition of 5' modified guanosine cap/ 3' poly-A tail ;
	7. No regulation of transcript stability;	Allows regulation of transcript stability;
Coupling of transcription and	8. Transcription & translation are tightly coupled ;	Nuclear membrane separates transcription and translation;

translation		
Regulation at translation level	9. Regulation at the translation level is less prominent ;	Regulation at the translation level is more prominent/ Regulation by translation initiation factors /negative translational control through repressor protein / cytoplasmic elongation of poly (A) tails / mRNA degradation;
Post-translational modification	10. Post-translational modifications rarely occur;	Post-translational modifications occur e.g. proteolysis, chemical modification, ubiquitination for degradation (name 1 example);
Gene amplification	11. No gene amplification;	Gene amplification (occurs to increase expression of a particular gene);

10(b) Explain how a deletion mutation in the gene for DNA polymerase III will prevent the occurrence of DNA replication. [6]

1. Deletion of nucleotides not in multiples of three will lead to a frameshift mutation;
2. Which result in change in codon sequence/ result in premature stop codon;
3. And hence different sequence of amino acids in the polypeptide/ resulting in truncated polypeptide;
4. Deletion of nucleotides in multiples of three will result in the deletion of essential amino acids (binding residue/ catalytic residue);
5. Ref. to change in bonds between R groups in polypeptide;
6. Resulting in change in 3D configuration of DNA polymerase III such that active site is no longer complementary in shape to the deoxyribonucleotides;
7. Hence DNA polymerase III will no longer be able to catalyse the condensation reaction/ phosphodiester bond between deoxyribonucleotides;
8. No formation of daughter DNA strand/ DNA molecule;

OR

9. Deletion of nucleotides in promoter;
10. Resulting in change in 3D conformation of promoter region
11. Not complementary to RNA polymerase/ cannot bind
12. No transcription and translation of DNA polymerase III;
13. Hence DNA polymerase III will no longer be able to catalyse the condensation reaction/ phosphodiester bond between deoxyribonucleotides;
14. No formation of daughter DNA strand/ DNA molecule;

@1m, max 6

10(c) Explain how biogeography supports the theory of evolution by natural selection. [6]

1. Biogeography is the geographical distribution of species (over time);
2. Discovered that species are not evenly distributed around the globe;
3. Closely related species tend to be located geographically closer together more than distantly related groups
4. Ref. to an example of distribution of closely related species e.g. Darwin's finches;
5. These findings are also possible evidence that organisms evolve from common ancestors;
6. They are located only in specific regions instead of being found wherever suitable habitats are located (as they are limited in their ability to colonise similar habitats in distant areas) due to geographical barriers;
7. Distantly related species in similar environments may appear similar due to convergent evolution;
8. Ref. to an example of convergent evolution e.g. flying squirrel and birds in rainforests;
9. These findings suggest that organisms are selected for their ability to adapt to a specific environment, which support theory of natural selection;
10. Ref. to an example of related species that are found far apart on different continents due to continental drift theory;

@1m, max 6